M5Camera

SKU:U017





!!! Extended use of the camera module is not recommended due to its tendency to overheat.



Filming in short bursts is recommended.

Description

M5Camera is an image recognition development board that integrates an ESP32 (4M Flash + 520K RAM + 4M PSRAM) chip and a 2MP camera (OV2640). It supports Wi-Fi image transmission and USB port debugging.

Features

- Designed based on ESP32
- Wi-Fi image transmission
- CP2104 USB TTL

Includes

- 1 x M5Camera
- 4 x LEGO Adapter
- 1 x USB Type-C Cable (20cm)

Specifications

Specification	Parameter
Flash	4M
PSRAM	4MB Quad
Image Sensor	OV2640
Max Resolution	2MP
Output Format	YUV (422/420) /YCbCr422, 8-bit compressed data, RGB565/555, 8-/10-bit Raw RGB data
FOV	65°
Product Size	40.0 x 48.0 x 11.0mm
Product Weight	17.0g
Package Size	75.0 x 45.0 x 30.0mm
Gross Weight	41.0g

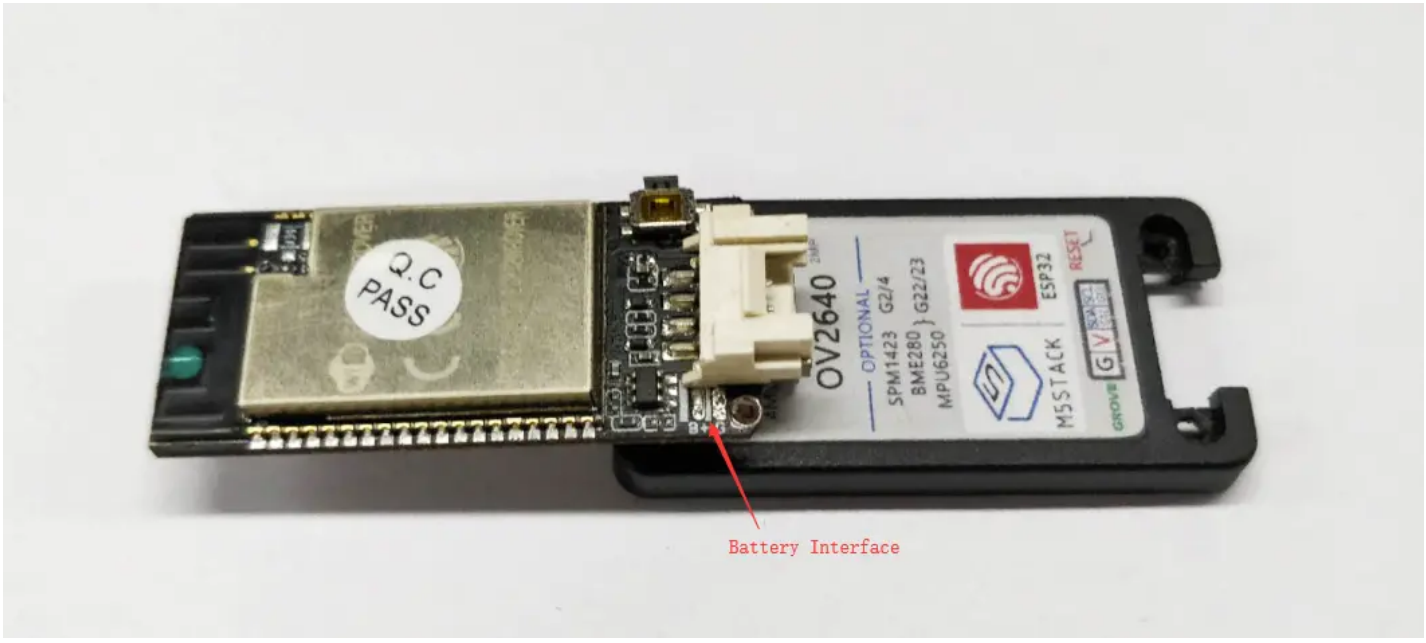
Learn

Hardware comes with pre-installed firmware, developed via ESP-IDF programming, running a Wi-Fi camera application. The default program outputs an image size of **600 x 800**, and you can optimize the program to output a larger size.

How is this program used?

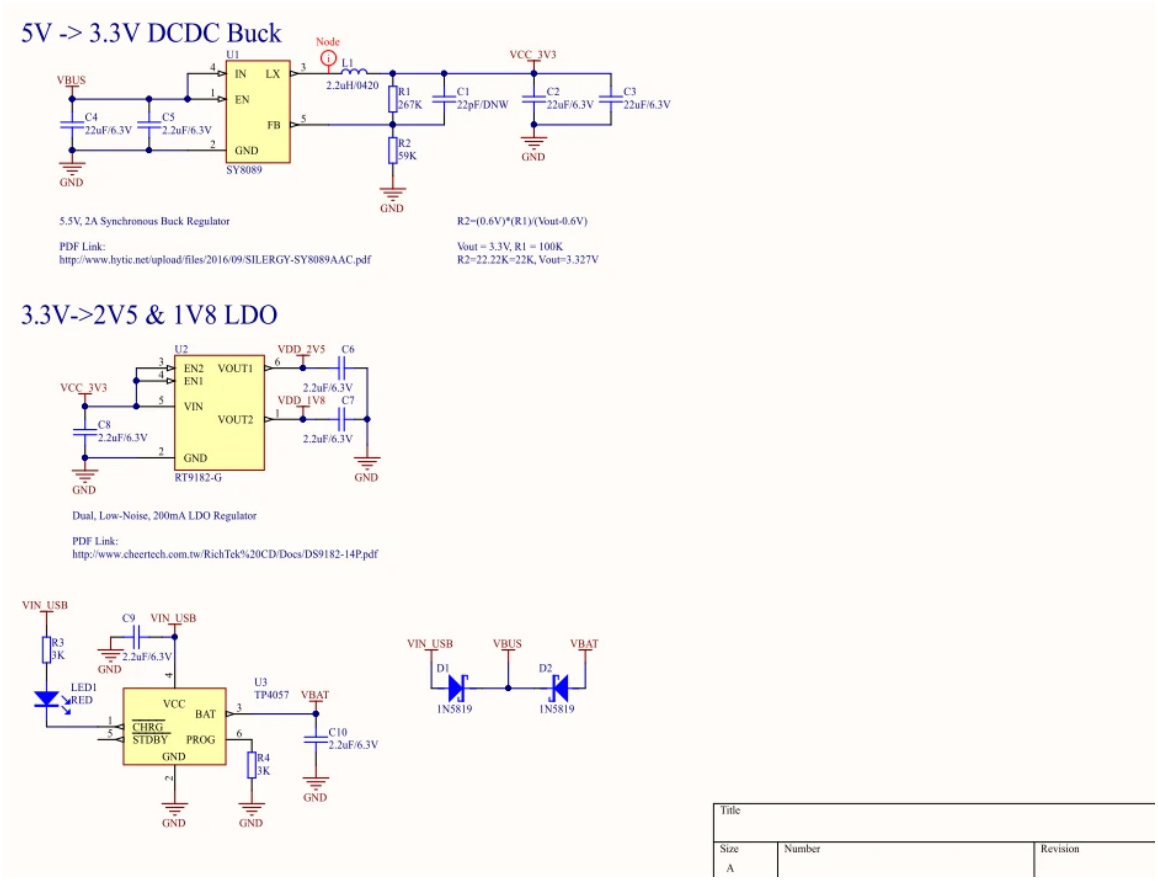
- Turn on your phone’ s Wi-Fi, scan and connect to an AP hotspot whose name starts with "m5stack-".
- Open the phone browser, visit 192.168.4.1, and enter the monitoring page to get real-time video feed.
- The video frame rate is about 5-6 frames per second.



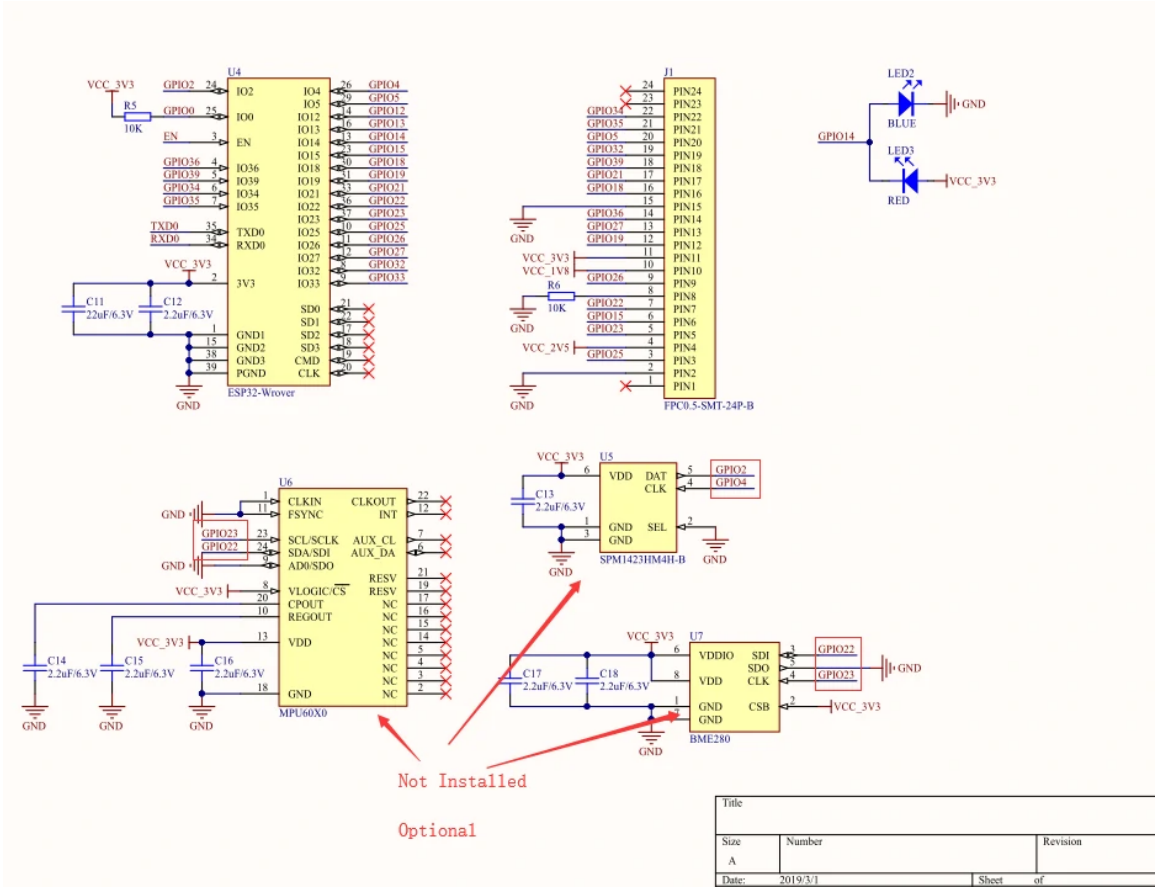


Schematics

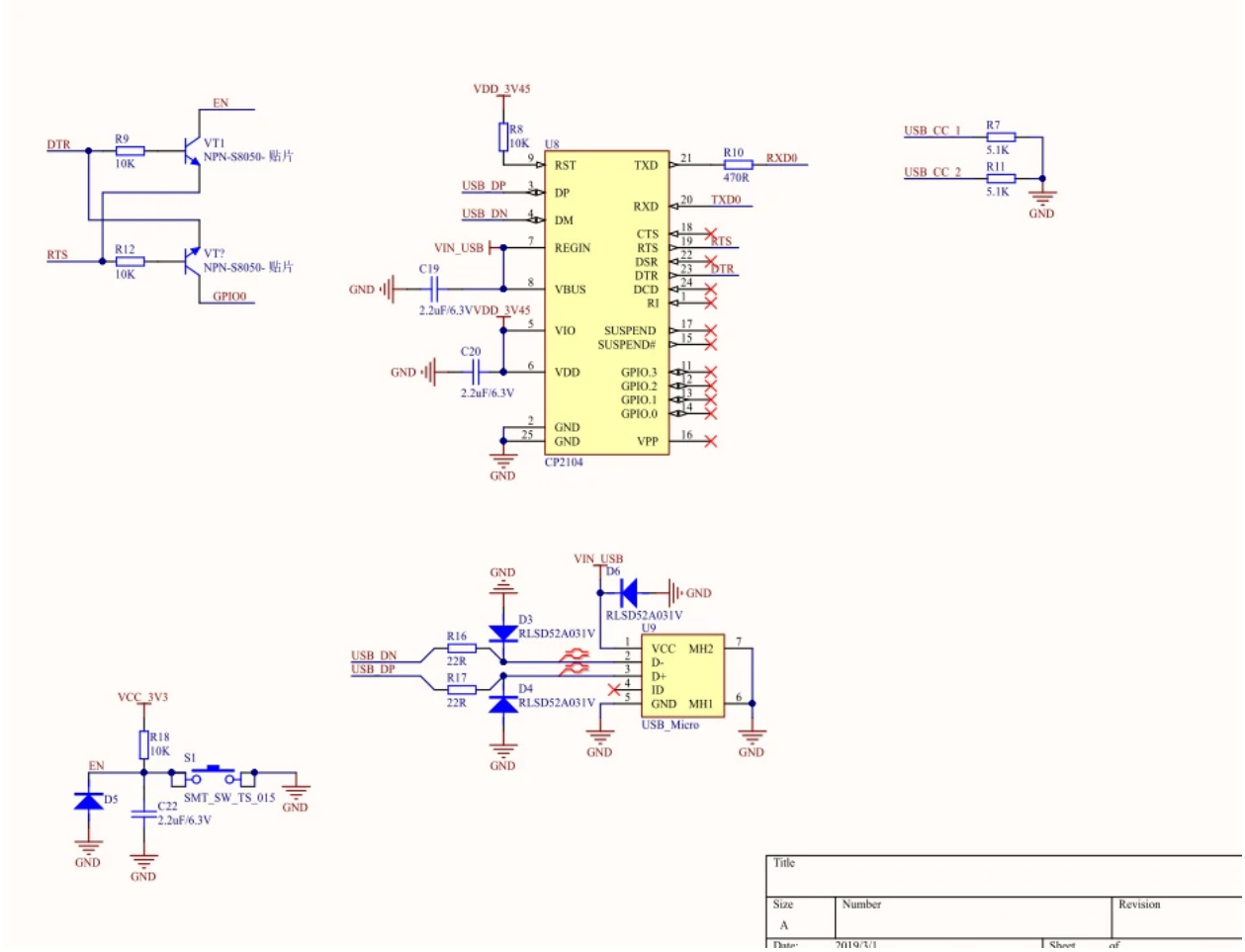
Power Circuit



Chip Peripheral Circuit



USB to Serial Circuit



PinMap

Camera Driver Chip OV2640 Interface

Interface	Camera Pin	M5Camera
SCCB Clock	SIOC	G23
SCCB Data	SIOD	G22
System Clock	XCLK	G27
Vertical Sync	VSYNC	G25
Horizontal Reference	HREF	G26
Pixel Clock	PCLK	G21
Pixel Data Bit 0	D2	G32
Pixel Data Bit 1	D3	G35
Pixel Data Bit 2	D4	G34
Pixel Data Bit 3	D5	G5
Pixel Data Bit 4	D6	G39
Pixel Data Bit 5	D7	G18
Pixel Data Bit 6	D8	G36
Pixel Data Bit 7	D9	G19
Camera Reset	RESET	G15
Camera Power Down	PWDN	/
Power Supply 3.3V	3V3	3V3
Ground	GND	GND

The **PIN8 (PDWN)** pin of the OV2640 chip is an enable pin. In this board, it is enabled by connecting to ground via a 12KΩ pull-down resistor, entering working mode. When the PIN8 (PDWN) pin is pulled high, it will enter **Camera Power Down** mode.

HY2.0-4P Interface

HY2.0-4P	M5Camera
SCL	IO13
SDA	IO4
5V	5V
GND	GND

LED Interface

LED	M5Camera
LED_Pin	IO14

The following are PCB reserved interfaces, with components not actually soldered.

BME280 Interface

BMP280	M5Camera
SCL	IO23
SDA	IO22

MPU6050 Interface

MPU6050	M5Camera
SCL	IO23
SDA	IO22

MIC (SPM1423) Interface

MIC (SPM1423)	M5Camera
CLK	IO4
DATA	IO2



Datasheets

- ESP32
- OV2640

Softwares

Internal Firmware (Wi-Fi Image Transmission)

- **M5Camera**

Arduino

- M5Camera-Arduino

ESP-IDF

- M5Camera
- Face Recognition
- Serial Communication - M5Camera
- Serial Communication - M5Core

- QRcode Recognition
- MPU6050

EasyLoader

Easyloader	Download	Note
M5Camera AP Firmware Easyloader	download	/

Video

- M5Camera Application — M5Camera and M5Core Image Transmission

M5Camera.mp4

Product Comparison

- View M5Camera Series / Product Differences